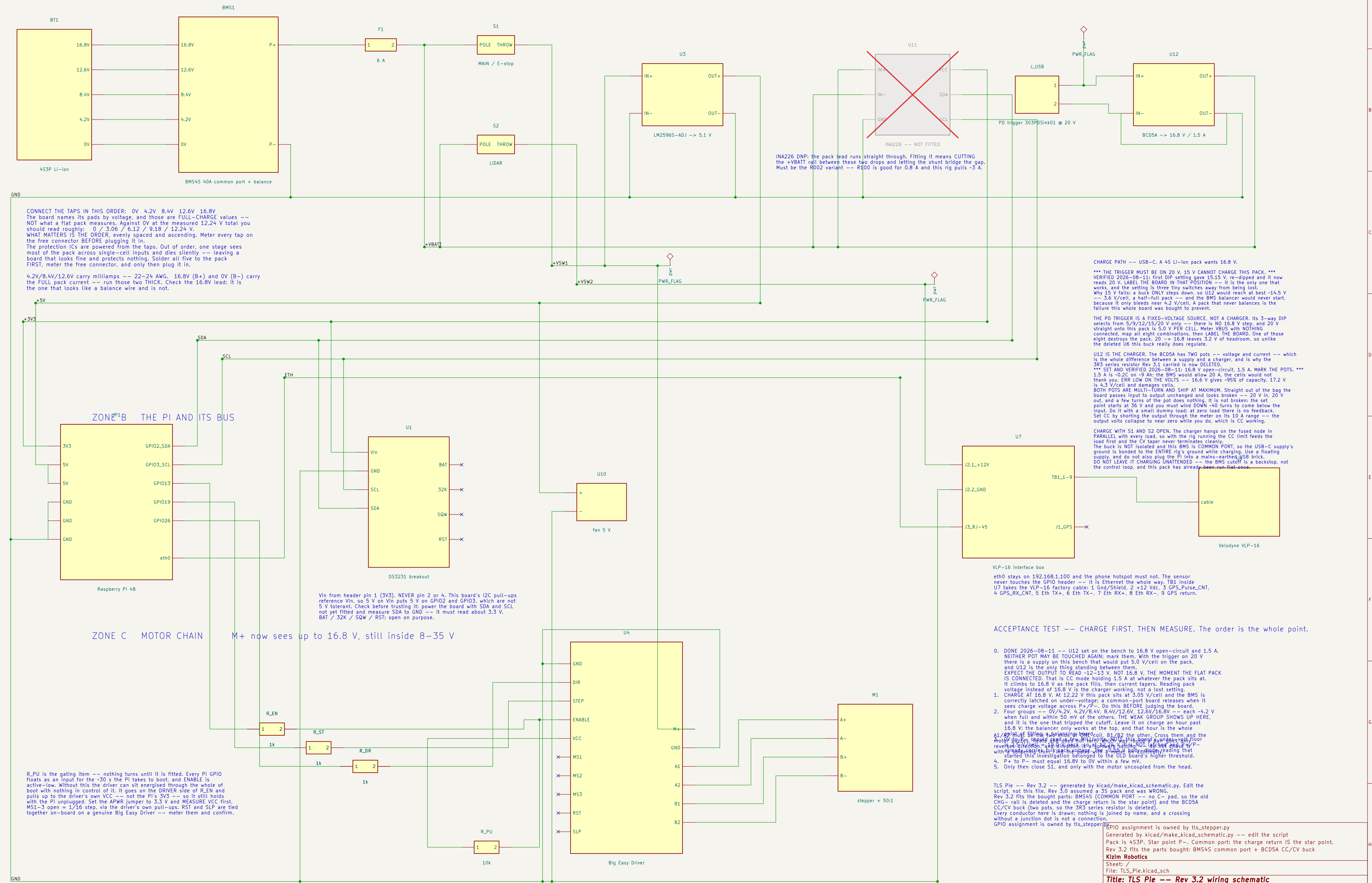


ZONE A POWER CHAIN 4S3P pack, common-port balancing BMS, USB-C charge

S1 and S2 both hang off the fused node, in PARALLEL. S1 does NOT switch the lidar -- opening S1 kills the Pi and the motor and leaves the VLP-16 spinning on S2.

*** NEW ON 4S: S2 HANDS THE VLP-16 UP TO 16.8 V ***
On the old 3S assumption that leg never passed 12.6 V. CHECK THE SENSOR'S OWN INPUT RANGE before S2 is ever closed. If it will not take 16.8 V that leg needs a regulator -- the one place going to 4S makes things worse.



ACCEPTANCE TEST -- CHARGE FIRST, THEN MEASURE. The order is the whole point.

- 0 DONE 2026-08-11 -- U12 set again on the bench to 16.8 V open-circuit and 1.5 A.
1 NEITHER POT MAY BE TOUCHED AGAIN; mark them. With the trigger on 20 V
2 the supply the pack sits at 16.8 V, the pot 0.0 V/cell on the pack,
3 and U12 is the only thing standing between them and each other.
4 EXPECT THE OUTPUT TO READ -12~13 V. PUT 16.8V ON THE MOMENT THE FLAT PACK IS
5 CONNECTED. That is CC made holding 1.5 A at whatever the pack sits at
6 and 16.8 V as the charger voltage. The pack sits at 16.8 V, the pot 0.0 V/cell
7 voltage instead of 16.8 V is the charger working, not a lost setting.
8 CHARGE AT 16.9 V. At 12.22 V this pack sits at 3.05 V/cell and the BMS is
9 charging it under load. Voltage is 16.8 V, the pot 0.0 V/cell, the balancer
10 sees charge voltage across P. NO DO BEFORE JUDGING! It takes time to
11 even up -- 0V/4.2V, 4.2V/8.4V, 8.4V/12.6V, 12.6V/16.8V -- each +4.2 V.
12 When full and within 50 mV of the others. THE WEAK GROUP SHOWS UP HERE.
13 Four groups in one that trip out. One group trips out at 16.8 V, the rest
14 16.8 V; the balancer only works at the top, and that hour is the whole
15 AT 16.8V still being a weak group. On the old cell, B1/B2 the other. Cross them and the motor
16 does not stop. They are good enough to work. No way to tell if they are with floor
17 or above floor. If they are with floor, they are good enough to work. If they are
18 with above floor, they are good enough to work. If they are with below floor,
19 they are good enough to work. If they are with above floor, they are good enough to work.
20 started this investigation belonged to the OLD board's higher threshold.
21 To get to equal 16.8 V, the pot must be set to 0.0 V/cell. The balancer
22 only works at the top, and that hour is the whole.
- 1 2
- 3 4
- 5 Only then close S1, and only with the motor uncoupled from the head.

TIS file -- Rev. 3.2 -- generated by kicad/make_kicad_schematic.py. Edit the script, not this file. Rev. 3.0 assumed a 35 pack and was WRONG.
 Rev. 3.2 fits the bought parts: BM545 (COMMON PORT --- no C-pad, so the o CHG- rail is deleted and the charge return is the star point) and the BCD5A CC/CV buck (two pots, so the 3R3 series resistor is deleted).
 Every conductor here is drawn: nothing is joined by name, and a crossing without a junction dot is not a connection.
 GPIO assignment is owned by tIs_stepper.py. GPIO assignment is owned by the

P1PIO assignment is owned by tls_stepper.py
 Generated by kicad/make_kicad_schematic.py -- edit the script
 Pack is 453P. Star point P--. Common port: the charge return IS the star point.
 Rev 3.2 fits the parts bought: BMS45 common port + BCD54 CC/CV buck
Kizim Robotics
 Sheet: /
 File: TLS_Pie_kicad_sch

Title: TLS Pie --- Rev 3.2 wiring schematic		
Size: A2	Date: 2026-08-11	Rev: 3.
KiCad E.D.A. 10.0.5		Id: 1/1